

MediTrust

Sprint_3_Backlog

Group7

The goal of Sprint 3 is to complete the remaining features of the MediTrust system, improve the user interface and usability, and ensure that the system functions as a fully integrated end-to-end application.

In Sprint 2, we successfully integrated a machine learning model for cardiovascular risk prediction and connected it with the frontend and backend system. However, the system still lacked explainability, proper input validation, and a refined user interface that would make it suitable for real-world use.

In this sprint, we focused on enhancing the system by adding explainable AI capabilities using SHAP, improving the user interface and user experience, implementing proper validation for clinical inputs, enhancing authentication features such as forgot password functionality, and ensuring that all system components work together seamlessly.

The system should now allow users to enter patient clinical parameters, receive a predicted cardiovascular risk level, understand the reasoning behind the prediction, and interact with a more professional and user-friendly interface.

User Stories

1. Explainable AI

As a clinician, I want to understand how each feature contributes to the prediction so that I can trust the system output.

2. Improved UI/UX

As a user, I want a clean and professional interface so that the system is easy to use and visually appealing.

3. Input Validation

As a system, I want to validate user inputs based on clinical standards so that incorrect data is not processed.

4. Authentication Enhancement

As a user, I want standard login features like forgot password so that I can securely access my account.

5. System Integration

As a developer, I want all components (frontend, backend, ML model) to work together seamlessly.

Task ID	Task Description	Assigned To	Priority	Est. Time	Status
T1	Create feature branches and plan Sprint 3 tasks.	Ravi	High	1 hr	Completed
T2	Enhance prediction API with explainability.	Ravi	High	4 hrs	Completed
T3	Integrate SHAP for model explainability.	Ravi	High	5 hrs	Completed
T4	Implement and verify prediction logging.	Ravi	High	3 hrs	Completed
T5	Enhance login system with standard features.	Uday	Medium	4 hrs	Completed
T6	Implement forgot password functionality.	Uday	Medium	3 hrs	Completed
T7	Connect frontend with backend prediction API.	Uday	High	3 hrs	Completed
T8	Enhance assessment form UI/UX.	Uday & Ravi	High	5 hrs	Completed
T9	Improve input validation and error handling.	Ravi	High	3 hrs	Completed
T10	Design prediction result page with explainability.	Ravi	High	4 hrs	Completed
T11	Perform end-to-end system integration testing.	Uday	High	3 hrs	Completed
T12	Update project board and prepare Sprint 3 documentation.	Ravi	Medium	4 hrs	Completed

All planned tasks for Sprint 3 were completed successfully.

At the end of this sprint, the system now supports:

- Explainable AI using SHAP
- Improved user interface and usability
- Clinical input validation
- Enhanced authentication features
- Full end-to-end workflow integration

Project management GitHub Board link: <https://github.com/users/RavindraSSK/projects/7>